

Foundations (pre-midterm, still needed)

RREF row-reduce (Gaussian elim.); leading 1's = *pivots*. $\text{rank } A = \# \text{pivots} = r$.

Solve $A\mathbf{x} = \mathbf{b}$ solvable $\iff \mathbf{b} \in \text{Col}(A) \iff \text{rank}[A|\mathbf{b}] = \text{rank } A$.
General sol = particular + nullspace.

Basis of $N(A)$ set free vars = 1 one at a time, solve pivots $\Rightarrow$ special solutions.

Lin. indep. only $c_1\mathbf{v}_1 + \dots = \mathbf{0}$ is trivial; square: $\iff \det \neq 0 \iff$ full rank.

Four subspaces $\text{Col}(A) \subset \mathbb{R}^m$ ($\dim r$), $N(A) \subset \mathbb{R}^n$ ($n-r$),
 $\text{Row}(A) = \text{Col}(A^T)$ (r), $N(A^T)$ ($m-r$).

Rank-nullity $\text{rank } A + \dim N(A) = n$ ($\# \text{cols}$).

Orthogonality $\text{Row}(A) \perp N(A)$; $\text{Col}(A) \perp N(A^T)$.

Invertible ($n \times n$): $\det \neq 0 \iff \text{rank } n \iff N = \{\mathbf{0}\} \iff \text{cols indep} \iff 0 \text{ not an eigenvalue.}$

Invertible Matrix Theorem (square A , TFAE)

A^{-1} exists $\iff \det A \neq 0 \iff \text{rank } A = n \iff n$ pivots $\iff$ columns (rows) independent $\iff$ columns span $\mathbb{R}^n \iff A\mathbf{x} = \mathbf{0}$ only at $\mathbf{x} = \mathbf{0} \iff A\mathbf{x} = \mathbf{b}$ has a unique solution $\iff 0$ is not an eigenvalue $\iff A^T A$ invertible.

Determinants

2×2 $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$. 3×3 Sarrus / cofactor.

Cofactor expand $\det A = \sum_j a_{ij} C_{ij}$, $C_{ij} = (-1)^{i+j} M_{ij}$ (any row/col; pick one with zeros).

Triangular $\det = \prod(\text{diagonal})$.

Row ops swap $\Rightarrow \times(-1)$; scale row by $k \Rightarrow \times k$; add multiple of a row $\Rightarrow$ unchanged.

Rules $\det(AB) = \det A \det B$; $\det A^T = \det A$; $\det(A^{-1}) = 1/\det A$;
 $\det(cA) = c^n \det A$.

Singular $\det A = 0 \iff A$ not invertible $\iff$ dependent columns.

Inverse $A^{-1} = \frac{1}{\det A} \text{adj}(A)$, $\text{adj} = [C_{ij}]^T$.

Link to eig. $\det A = \prod \lambda_i$. Ex $\det B = 4$ (3×3): $\det(2B^{-1}) = 2^3 \cdot \frac{1}{4} = 2$.

Trace & Determinant identities

$\text{tr}(A+B) = \text{tr } A + \text{tr } B$; $\text{tr}(AB) = \text{tr}(BA)$ (cyclic).

$\text{tr } A = \sum \lambda_i$; $\det A = \prod \lambda_i = \pm \prod(\text{pivots})$.

$\det(AB) = \det A \det B$; $|\det A| = \text{volume scaling factor of } A$.

Eigenvalues & Eigenvectors (MAIN FOCUS)

Definition $A\mathbf{v} = \lambda\mathbf{v}$, $\mathbf{v} \neq \mathbf{0}$.

Find λ solve $\det(A - \lambda I) = 0$ (characteristic polynomial, degree n).

Find $\mathbf{v}$ $E_\lambda = N(A - \lambda I)$ = the eigenspace; row-reduce $A - \lambda I$, take special solutions for a basis.

Trace & det $\sum \lambda_i = \text{tr } A$, $\prod \lambda_i = \det A$.

Multiplicities geom mult = $\dim E_\lambda = n - \text{rank}(A - \lambda I)$; always $1 \leq$ geom $\leq$ alg.

Triangular eigenvalues = diagonal entries.

Shifts/powers $A^k : \lambda^k$; $A^{-1} : 1/\lambda$; $A + cI : \lambda + c$; eigvecs unchanged.
 $\text{tr}(A^2) = \sum \lambda_i^2$.

Diagonalize $A = S\Lambda S^{-1}$, $S = [\mathbf{v}_1 \dots \mathbf{v}_n]$, $\Lambda = \text{diag}(\lambda_i)$.

A diagonalizable $\iff n$ indep. eigvecs $\iff \text{geom} = \text{alg } \forall \lambda$.

Then $A^k = S\Lambda^k S^{-1}$. Distinct $\lambda \Rightarrow$ auto-diagonalizable.

Similar $B = M^{-1}AM \Rightarrow$ same char. poly, λ 's, tr, det.

Real matrix complex λ come in conjugate pairs $a \pm bi$.

Cayley-Hamilton A satisfies its own char. poly: $p(A) = 0$. 2×2 :
 $A^2 - (\text{tr } A)A + (\det A)I = 0 \Rightarrow A^{-1} = \frac{(\text{tr } A)I - A}{\det A}$.

Worked eigen $\rightarrow$ diagonalize

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: $\det(A - \lambda I) = (2 - \lambda)^2 - 1 = (\lambda - 1)(\lambda - 3)$.

$\lambda=1$: $A - I = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \Rightarrow \mathbf{v} = (1, -1)$. $\lambda=3$: $\mathbf{v} = (1, 1)$.

$S = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$, $\Lambda = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$. Symmetric $\Rightarrow$ orthonormalize: $A = Q\Lambda Q^T$,
 $Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$.

Special matrices' eigenvalues

Projection $\lambda \in \{0, 1\}$. **Reflection** $\lambda = \pm 1$.

Orthogonal Q $|\lambda| = 1$. **Rotation** $e^{\pm i\theta} = \cos \theta \pm i \sin \theta$.

Nilpotent ($A^k = 0$) all $\lambda = 0$. **Idempotent** ($A^2 = A$) $\lambda \in \{0, 1\}$.

Markov (cols sum 1) has $\lambda = 1$. **Singular** has $\lambda = 0$.

Rotation ex $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$: $\lambda = \pm i$, eigvecs $(1, \mp i)$.

Defective (NOT diagonalizable) example

$A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$: $\lambda = 3$ with alg mult 2. $A - 3I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has rank 1 $\Rightarrow$ geom mult = $2 - 1 = 1 < 2$. Too few eigenvectors $\Rightarrow$ **not** diagonalizable.

Orthogonal Projection & Least Squares

Onto a line $\mathbf{a}$ $\mathbf{x}_\parallel = \frac{\mathbf{a}^T \mathbf{b}}{\mathbf{a}^T \mathbf{a}} \mathbf{a}$; matrix $P = \frac{\mathbf{a} \mathbf{a}^T}{\mathbf{a}^T \mathbf{a}}$.

Onto $\text{Col}(A)$ (indep cols): $P = A(A^T A)^{-1} A^T$, projection $\hat{\mathbf{b}} = P\mathbf{b}$.

P facts $P^2 = P$, $P^T = P$, eigenvalues 0/1; $I - P$ projects onto $N(A^T)$.

Least squares solve $A^T A \hat{\mathbf{x}} = A^T \mathbf{b} \Rightarrow \hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$ minimizes $\|A\mathbf{x} - \mathbf{b}\|^2$; error $\mathbf{b} - A\hat{\mathbf{x}} \perp \text{Col}(A)$.

Worked least squares (line fit)

Fit $b = C + Dt$ to $(0, 6), (1, 0), (2, 0)$: $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}$.

$A^T A = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}$, $A^T \mathbf{b} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} \Rightarrow \hat{\mathbf{x}} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}$: best line $b = 5 - 3t$, projection $A\hat{\mathbf{x}} = (5, 2, -1)$.

Gram-Schmidt & QR Decomposition

Orthonormalize $\mathbf{v}_k = \mathbf{a}_k - \sum_{j < k} (\mathbf{q}_j^T \mathbf{a}_k) \mathbf{q}_j$, then $\mathbf{q}_k = \mathbf{v}_k / \|\mathbf{v}_k\|$.

QR $A = QR$: Q has orthonormal columns ($Q^T Q = I$), R upper-triangular,

$R_{kk} = \|\mathbf{v}_k\|$, $R_{jk} = \mathbf{q}_j^T \mathbf{a}_k$ ($j < k$). (Keep $R_{kk} > 0$.)

Orthogonal matrix $Q^T Q = I \Rightarrow Q^{-1} = Q^T$, preserves length & angle, $\det Q = \pm 1$.

Use QR makes least squares easy: $R\hat{\mathbf{x}} = Q^T \mathbf{b}$.

Worked Gram-Schmidt / QR

$\mathbf{a}_1 = (1, 1, 0)$, $\mathbf{a}_2 = (1, 0, 1)$: $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$.

$\mathbf{v}_2 = \mathbf{a}_2 - (\mathbf{q}_1^T \mathbf{a}_2) \mathbf{q}_1 = (\frac{1}{2}, -\frac{1}{2}, 1) \Rightarrow \mathbf{q}_2 = \frac{1}{\sqrt{6}}(1, -1, 2)$.

$R = \begin{pmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{pmatrix}$, $Q = [\mathbf{q}_1 \ \mathbf{q}_2]$.

Symmetric Matrices & Spectral Theorem

$A = A^T \Rightarrow$ real λ , eigenvectors of distinct λ are orthogonal.

Spectral / orthogonal diag. $A = Q\Lambda Q^T$, Q orthogonal ($Q^T Q = I$), columns = orthonormal eigenvectors. ($= \sum \lambda_i \mathbf{q}_i \mathbf{q}_i^T$.)

Recipe per λ : basis of E_λ , then Gram-Schmidt within it; normalize all.

Definiteness & Rayleigh Quotient

For symmetric A (let λ be its eigenvalues):

Pos. definite $\mathbf{x}^T A \mathbf{x} > 0 \ \forall \mathbf{x} \neq \mathbf{0} \iff$ all $\lambda > 0 \iff$ all pivots $> 0 \iff$ all leading principal minors $> 0 \iff A = R^T R$ (R full rank).

Pos. semidef. $\lambda \geq 0$; **neg. def.** $\lambda < 0$; **indefinite** mixed-sign λ .

2×2 $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ PD $\iff a > 0$ & $ac - b^2 > 0$.

Rayleigh $R(\mathbf{x}) = \frac{\mathbf{x}^T A \mathbf{x}}{\mathbf{x}^T \mathbf{x}} \in [\lambda_{\min}, \lambda_{\max}]$; $\max = \lambda_{\max}$, $\min = \lambda_{\min}$ attained at the corresponding eigenvector.

Worked $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$: minors $2 > 0$, $\det 3 > 0 \Rightarrow$ PD; check $\lambda = 1, 3 > 0$. $R(\mathbf{x}) \in [1, 3]$.

Pivots vs Eigenvalues vs Singular values

Pivots from elimination; signs $\rightarrow$ det & definiteness (symmetric).

Eigenvalues $A\mathbf{x} = \lambda\mathbf{x}$; can be complex/negative.

Singular values $\sigma_i = \sqrt{\lambda_i(A^T A)} \geq 0$ always real.

Symmetric PD $\Rightarrow$ all three positive (but generally unequal); symmetric $\Rightarrow \sigma_i = |\lambda_i|$.

Singular Value Decomposition (SVD)

Form $A = U \Sigma V^T$ ($m \times n$): U ($m \times m$), V ($n \times n$) orthogonal; Σ diagonal $\sigma_1 \geq \dots \geq 0$.
Compute V = orthonormal eigvecs of $A^T A$; $\sigma_i = \sqrt{\lambda_i(A^T A)}$; U = eigvecs of AA^T , or $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$.
Facts $\#\{\sigma_i > 0\} = \text{rank } A$; $\sigma_{\max} = \|A\|_2$; $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$.
Pseudoinverse $A^+ = V \Sigma^+ U^T$ (least-squares / min-norm).
Geometry A maps the unit sphere to an ellipsoid with semi-axis *lengths* σ_i along directions $\mathbf{u}_i$.
Norms $\|A\|_2 = \sigma_{\max}$; $\|A\|_F = \sqrt{\sum a_{ij}^2} = \sqrt{\sum \sigma_i^2}$. **Condition** $\kappa_2 = \sigma_{\max}/\sigma_{\min}$.

Worked SVD: ellipse axes

$B = \begin{pmatrix} 2 & 2 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$: $B^T B = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$, $\lambda = 9, 1 \Rightarrow \sigma = 3, 1$.
 $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1) \rightarrow \mathbf{u}_1 = \frac{B\mathbf{v}_1}{3} = \frac{1}{3\sqrt{2}}(4, 1, 1)$ (long axis, len 3);
 $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1) \rightarrow \mathbf{u}_2 = \frac{1}{\sqrt{2}}(0, 1, -1)$ (short axis, len 1).

Linear Maps, Matrix Rep. & Change of Basis

Matrix of L in basis $\mathcal{B}$: column j of $[L]_{\mathcal{B}}$ = coords of $L(\mathbf{b}_j)$ in $\mathcal{B}$.
Coords $[\mathbf{v}]_{\mathcal{B}} = P_{\mathcal{B} \leftarrow \mathcal{C}}[\mathbf{v}]_{\mathcal{C}}$, where $P_{\mathcal{B} \leftarrow \mathcal{C}}$ has columns = $\mathcal{C}$ -vectors written in $\mathcal{B}$.
Similarity $[L]_{\mathcal{C}} = P^{-1}[L]_{\mathcal{B}}P$, $P = P_{\mathcal{B} \leftarrow \mathcal{C}}$.
Eigen of L alg/geom mult as for matrices; L diagonalizable $\iff$ eigenvectors form a basis.

Block Matrices

Block-triangular $\det \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det A \det D$.
Schur complement (A invertible): $\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \det(D - CA^{-1}B)$ (block Gaussian elim).
Use simplify big det/inverse by eliminating a block; eigenvectors often $\begin{pmatrix} \mathbf{v} \\ \alpha \mathbf{v} \end{pmatrix}$.

Vector Spaces & Subspaces

Subspace test contains $\mathbf{0}$, closed under $+$ and scalar mult.
Basis independent + spanning; **dim** = #basis vectors.
Common dims $\mathbb{P}_n$ ($\deg \leq n$): $n+1$; $M_{m \times n}$: mn ; symmetric $n \times n$: $\frac{n(n+1)}{2}$; trace-zero 2×2 : 3.
Coordinates pick basis $\Rightarrow$ work with coordinate vectors.

Inner Product, Norm, Orthogonality

$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y}$, $\|\mathbf{x}\| = \sqrt{\mathbf{x}^T \mathbf{x}}$, $\cos \theta = \frac{\mathbf{x}^T \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$.
Cauchy–Schwarz $|\mathbf{x}^T \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$.
Orthonormal $\mathbf{q}_i^T \mathbf{q}_j = \delta_{ij}$. $\perp$ **complement** $(\text{Col } A)^\perp = \text{N}(A^T)$.

Quick Reference

0 is eigenvalue $\iff \det A = 0 \iff$ singular. P projection $\Rightarrow \lambda \in \{0, 1\}$.
Orthogonal $Q \Rightarrow |\lambda| = 1$, $\det = \pm 1$. Symmetric $\Rightarrow \lambda \in \mathbb{R}$, $Q \Lambda Q^T$.
 $\sigma_i = \sqrt{\lambda_i(A^T A)} \geq 0$ always real. Diagonalizable $\neq$ invertible (independent).

Exam playbook (from 2025 final)

1. Basics: $\det = \prod \lambda$, $\text{tr} = \sum \lambda$, dependence $\iff \det = 0$. 2. Char. poly via cofactor; singular $\iff \det = 0$; complex eigvecs. 3. Spectral $Q \Lambda Q^T$; indefinite test; $\{p(A)\}$ basis (Cayley–Hamilton). 4. σ of A ; QR of $[A \ \mathbf{b}]$; projection P ; least squares $\hat{\mathbf{x}}$; ellipse via SVD. 5. Block det/inverse; eigvecs; positive-definite condition. 6. Map of L in a basis; alg/geom mult; diagonalizable?; change of basis.